

Field and Galois Theory

1.

- (a) Give an example of a tower of field extensions $F \supset L \supset K$ such that L/K is Galois and F/L is Galois, but F/K is not Galois.
- (b) Give an example of a primitive element in $\mathbb{Q}(\sqrt{2}, \sqrt{3})$. Prove that the element you give is indeed a primitive element. Find the minimal polynomial of the element you find.

Solution 1.

- (a) Consider $K = \mathbb{Q}$, $L = \mathbb{Q}(\sqrt{2})$, and $F = \mathbb{Q}(\sqrt[4]{2})$. Then $F \supset L \supset K$, and L/K is Galois since it's the splitting field of the separable polynomial $x^2 - 2 \in \mathbb{Q}[x]$, and F/L is Galois since it's the splitting field of the separable polynomial $x^2 - \sqrt{2} \in \mathbb{Q}(\sqrt{2})[x]$, but F/K is not Galois since it is not a splitting field (it doesn't contain the roots $\pm i\sqrt[4]{2}$ of the minimal polynomial of $\sqrt[4]{2}$, $m(x) = x^4 - 2 = (x - \sqrt[4]{2})(x + \sqrt[4]{2})(x - i\sqrt[4]{2})(x + i\sqrt[4]{2})$).
- (b) $\sqrt{2} + \sqrt{3}$ is a primitive element in $\mathbb{Q}(\sqrt{2}, \sqrt{3})$. Its minimal polynomial is

$$m(x) = (x - (\sqrt{2} + \sqrt{3}))(x - (\sqrt{2} - \sqrt{3}))(x + (\sqrt{2} + \sqrt{3}))(x + (\sqrt{2} - \sqrt{3})) = x^4 - 10x^2 + 1.$$

Therefore, $\mathbb{Q}(\sqrt{2} + \sqrt{3})/\mathbb{Q}$ is a degree 4 extension, and $\mathbb{Q}(\sqrt{2} + \sqrt{3}) \subseteq \mathbb{Q}(\sqrt{2}, \sqrt{3})$. But $\mathbb{Q} \subset \mathbb{Q}(\sqrt{2}) \subset \mathbb{Q}(\sqrt{2}, \sqrt{3})$ and $[\mathbb{Q}(\sqrt{2}) : \mathbb{Q}] = 2$ and $[\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}(\sqrt{2})] \leq 2$, so $[\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}] \leq 4$. Thus, we must have equality, so $\mathbb{Q}(\sqrt{2} + \sqrt{3}) = \mathbb{Q}(\sqrt{2}, \sqrt{3})$.

* 2.

Let $\mathbb{F}_p$ denote the field with p elements.

- (a) What is the factorization of the polynomial $x^4 + 1$ into irreducible factors in $\mathbb{F}_3[x]$?
- (b) Give an example of an element $a \in \mathbb{F}_5$ such that $x^4 - a$ is irreducible in $\mathbb{F}_5[x]$, and prove that $x^4 - a$ is irreducible with this choice of a .
- (c) Prove that for any nonzero choice of $a \in \mathbb{F}_{103}$, the polynomial $x^4 - a$ factors either as a product of two irreducible quadratic polynomials, or as a product of two linear polynomials and an irreducible quadratic.

Solution 2.

- (a) Observe that $x^4 + 1$ has no roots in $\mathbb{F}_3$, so it has no linear factors over $\mathbb{F}_3[x]$. Therefore, if it factors it must be as $(x^2 + ax + b)(x^2 + cx + d)$ for some $a, b, c, d \in \mathbb{F}_3$. Multiplying out, we see that we must have $a + c = 0$, $ac + b + d = 0$, and $bd = 1$. The last equation only has solutions $b = d = 1$ and $b = d = 2$. The former gives no solutions to the remaining equations, and the latter gives the solutions $a = 1, c = 2$ and $a = 2, c = 1$. Thus, $x^4 + 1 = (x^2 + x + 2)(x^2 + 2x + 2)$ over $\mathbb{F}_3[x]$.
- (b) Consider $a = 2$. Then $x^4 - 2$ is irreducible in $\mathbb{F}_5[x]$. Indeed, it has no linear factors since $\alpha^4 \in \{0, 1\}$ for all $\alpha \in \mathbb{F}_5$. So, we just need to show that it has no quadratic factors $x^2 + bx + c$ with $b, c \in \mathbb{F}_5$. Observe that if $x^2 = -bx - c$, then $x^4 = (-b^3 + 2bc)x - b^2c + c^2$. So, we need to show that if $-b^3 + 2bc = 0$, then $-b^2c + c^2 \neq 2$. If $b = 0$, then it is clear

that $c^2 \neq 2$ for any choice of c . If $b \neq 0$, then the only possible solutions for $-b^3 + 2c = 0$ are $b \in \{1, 4\}, c = 3$ and $b = \{2, 3\}, c = 3$. In the first case, $-b^2c + c^2 = 0$, and in the second case, $-b^2c + c^2 = 3$. Therefore, $x^4 - 2$ is irreducible in $\mathbb{F}_5[x]$.

- (c) Since $103 \equiv 3 \pmod{4}$, it is a fact that every nonzero $a \in \mathbb{F}_{103}$ satisfies either $a = c^2$ or $-a = c^2$ for some $c \in \mathbb{F}_{103}$. If $a = c^2$, then $x^4 - a = (x^2 - c)(x^2 + c)$, but since $c \in \mathbb{F}_{103}$ we must have that $c = d^2$ or $-c = d^2$, so exactly one of these factors reduces into $(x - d)(x + d)$. Thus, if a is a square mod 103, then $x^4 - a$ factors into two linear factors and an irreducible quadratic. Otherwise, if $-a = c^2$, then $x^4 - a = (x^2 + cx + c)(x^2 - cx + c)$, which is a product of two irreducible quadratics.

3.

Let ζ be a primitive k th root of unity.

- (a) Prove that $K = \mathbb{Q}(\zeta)$ is a Galois extension of $\mathbb{Q}$.
 (b) Compute $\text{Gal}(K/\mathbb{Q})$.

Solution 3.

- (a) Let $\Phi_k(x)$ be the k th cyclotomic polynomial, which is defined iteratively by

$$\Phi_k(x) = \frac{x^k - 1}{\prod_{d|k, d < k} \Phi_d(x)}.$$

Then $\Phi_k(x)$ is the minimal polynomial of ζ since it is irreducible over $\mathbb{Q}$. The roots of $\Phi_k(x)$ are ζ^i for each $1 \leq i \leq k$ with $\gcd(k, i) = 1$. Therefore, $K = \mathbb{Q}(\zeta)/\mathbb{Q}$ is Galois since $\Phi_k(x)$ is separable and K contains all of the roots of $\Phi_k(x)$.

- (b) By part (a), each automorphism of $K/\mathbb{Q}$ is determined by $\zeta \mapsto \zeta^i$ for some $1 \leq i \leq k$ with $\gcd(k, i) = 1$. Therefore, $|\text{Gal}(K/\mathbb{Q})| = \varphi(k)$ and $\text{Gal}(K/\mathbb{Q}) = (\mathbb{Z}/k\mathbb{Z})^*$.

*** 4.**

- (a) Let a be a root of the polynomial $x^4 - 7 \in \mathbb{Q}[x]$. Show that $\mathbb{Q}(a)$ is not Galois over $\mathbb{Q}$.
 (b) Let F be a field of characteristic 5, and let b be a root of $x^4 - 7 \in F[x]$. Show that $F(b)$ is Galois over F .
 (c) Compute the Galois group $\text{Gal}(F(b)/F)$.

Solution 4.

- (a) The roots of $x^4 - 7$ are $\pm\sqrt[4]{7}$ and $\pm i\sqrt[4]{7}$. If $a = \pm\sqrt[4]{7}$, then $\mathbb{Q}(a)$ does not contain $i\sqrt[4]{7}$ so it is not a splitting field, and thus not Galois. If $a = \pm i\sqrt[4]{7}$, then $\mathbb{Q}(a)$ does not contain $\sqrt[4]{7}$, so again is not a splitting field so not Galois over $\mathbb{Q}$.
 (b) Since $x^4 - 7 \equiv x^4 - 2 \pmod{5}$, we get that the roots of this polynomial over F are $\pm b$ and $\pm ib$, where $i = \sqrt{-1}$. Again, since $\text{char}(F) = 5$, $i = \sqrt{4} = 2$, so the roots are $\pm b$ and $\pm 2b$, or, more simply, $b, 2b, 3b$, and $4b$. Since each of these is in $F(b)$, it is the splitting field of $x^4 - 7$, and since this polynomial is separable, $F(b)$ is Galois over F .

- (c) An automorphism $\sigma \in \text{Gal}(F(b)/F)$ is determined by which root it sends b to. Thus, the elements in $\text{Gal}(F(b)/F)$ are $\sigma_1 = \text{Id} : b \mapsto b$, $\sigma_2 : b \mapsto 2b$, $\sigma_3 : b \mapsto 3b$, and $\sigma_4 : b \mapsto 4b$. Observe that $\sigma_2^2 = \sigma_3^2 = \sigma_4$ and $\sigma_4^2 = \text{Id}$. Therefore, $\text{Gal}(F(b)/F) \cong (\mathbb{Z}/5\mathbb{Z})^*$ where the isomorphism is given by $\sigma_i \mapsto i$.

* 5.

This is a question about “biquadratic extensions” in two parts.

- (a) Let F/E be a degree-4 Galois extension, where E and F are fields of characteristic different from 2. Show that $\text{Gal}(F/E) \cong (\mathbb{Z}/2\mathbb{Z}) \times (\mathbb{Z}/2\mathbb{Z})$ if and only if there exist $x, y \in E$ such that $F = E(\sqrt{x}, \sqrt{y})$ and none of x, y, xy are squares in E .
- (b) (Slightly modified from original) Give an example of an infinite field E of characteristic 2 that is not algebraically closed but that has no Galois extension F/E with Galois group $(\mathbb{Z}/2\mathbb{Z}) \times (\mathbb{Z}/2\mathbb{Z})$.

Solution 5.

- (a) Suppose $F = E(\sqrt{x}, \sqrt{y})$, where none of x, y , and xy are squares in E . This is a degree 4 extension, since it has basis $1, \sqrt{x}, \sqrt{y}, \sqrt{xy}$. It has four easy-to-think-of automorphisms: the identity, σ which flips the sign of $\sqrt{x}$ and $\sqrt{xy}$, τ which flips the sign of $\sqrt{y}$ and $\sqrt{xy}$, and $\sigma\tau$ which flips the sign of $\sqrt{x}$ and $\sqrt{y}$. Since the automorphism group can have size at most 4, we know we’ve found all the automorphisms, so this is in fact a Galois extension.

On the other hand, suppose F/E is a degree 4 Galois extension with Galois group $(\mathbb{Z}/2\mathbb{Z}) \times (\mathbb{Z}/2\mathbb{Z})$. Then, by the Fundamental Theorem of Galois Theory, from the subgroup structure of $(\mathbb{Z}/2\mathbb{Z}) \times (\mathbb{Z}/2\mathbb{Z})$, we know that F has three intermediate extension, which are degree 2 over E . Since E is not characteristic 2, the quadratic formula tells us that any degree 2 extension is of the form $E(\sqrt{d})$ for some nonsquare d . So, we can say two of the intermediate extensions are $E(\sqrt{x})$ and $E(\sqrt{y})$. Since these are distinct extensions of E , $\sqrt{x} \notin E(\sqrt{y})$ and visa versa, so $E(\sqrt{x}, \sqrt{y})$ is a degree 4 extension of E sitting inside F , and thus in fact we have $F = E(\sqrt{x}, \sqrt{y})$.

- (b) Oops, this problem is really hard I think. I’m about 80% sure that $E = \mathbb{F}_2(t)$ is an example, but it seems like proving it is pretty ghastly.

6.

Suppose that $K/\mathbb{Q}$ is a degree-4 Galois extension with Galois group $\mathbb{Z}/4\mathbb{Z}$. Prove that $\sqrt{-1} \notin K$.

Solution 6.

Suppose that K is a degree 4 Galois extension, and that $\sqrt{-1} \in K$. Then $\mathbb{Q}(\sqrt{-1}) \subset K$ is a subfield. But also, if we embed K in $\mathbb{C}$, we can take the subfield of K fixed by complex conjugation. This subfield must be a degree-2 subextension of K , but it cannot be $\mathbb{Q}(\sqrt{-1})$, so K has two different degree-2 subextensions. This implies that $\text{Gal}(K/\mathbb{Q})$ has two different subgroups of order 2. But $\mathbb{Z}/4\mathbb{Z}$ has only one such subgroup, so $\text{Gal}(K/\mathbb{Q}) \neq \mathbb{Z}/4\mathbb{Z}$.

* 7.

For each of the following statements, either prove it or provide a counterexample.

- (a) Every matrix of finite order in $\mathrm{GL}_n(\mathbb{C})$ is diagonalizable.
- (b) Every matrix of finite order in $\mathrm{GL}_n(\mathbb{Q})$ is diagonalizable.
- (c) Every matrix in $\mathrm{GL}_n(\overline{\mathbb{F}_p})$ has finite order. (We write $\overline{\mathbb{F}_p}$ for the algebraic closure of the finite field $\mathbb{F}_p$.)
- (d) Every matrix of finite order in $\mathrm{GL}_n(\overline{\mathbb{F}_p})$ is diagonalizable.

Solution 7.

- (a) Suppose $A \in \mathrm{GL}_n(\mathbb{C})$ has finite order, so that $A^k = I$ for some k . Then we know the minimal polynomial of A divides $x^k - 1$. Now, over $\mathbb{C}$, $x^k - 1$ splits into k different linear factors, so the minimal polynomial also splits into different linear factors. Since the exponents on the linear factors of the minimal polynomial are the sizes of maximal Jordan blocks, this tells us that the size of a maximal Jordan block is 1, which is the same as saying A is diagonalizable.
- (b) Everyone's favorite counterexample:

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

- (c) Let $\mu_A(x)$ be the minimal polynomial of A . Let $\mathbb{F}_q \supset \mathbb{F}_p$ be the smallest extension containing all the roots of $\mu_A(x)$. Then all the roots of $\mu_A(x)$ are nonzero elements of $\mathbb{F}_q$, and so satisfy $\alpha^{q-1} = 1$. Since $\mu_A(x)$ factors over $\mathbb{F}_q$, write

$$\mu_A(x) = (x - \alpha_1)^{e_1} \cdots (x - \alpha_k)^{e_k}.$$

If all the e_i were 1 then we would just know that $\mu_A(x) \mid x^{q-1} - 1$, and we would be done, but if one of the $e_i > 1$ then we want to find another polynomial of the form $x^{\text{something}} - 1$ which μ_A divides. We know $x^{q-1} - 1$ has distinct roots, so if $e = \max_i(e_i)$, then $\mu_A(x) \mid (x^{q-1} - 1)^e$, but this isn't in the form we want yet. To get it in the desired form, we can instead choose the smallest power of p which is greater than e . Namely, set $m = \log_p(e)$, then we have $\mu_A(x) \mid (x^{q-1} - 1)^{p^m} = x^{p^m(q-1)} - 1$. Thus, $A^{p^m(q-1)} = I$. (I actually think this exponent might be optimal, but I'm not sure. Lemme know if you figure out one way or the other.)

- (d) The problem with our proof from part (a) is that we said $x^k - 1$ splits into *distinct* linear terms over $\mathbb{C}$, but that need not be true in $\overline{\mathbb{F}_p}$. For example, $x^p - 1$ definitely doesn't. So, take the example of

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

over $\overline{\mathbb{F}_2}$.

* 8.

Let L/K be a degree p Galois extension, where p is prime, and let σ be a nontrivial element of $\text{Gal}(L/K)$. Then σ is a linear transformation of the K -vector space L .

- (a) What is the characteristic polynomial of σ ?
- (b) Write L^0 for the subspace of L consisting of elements with trace 0. Show that σ sends L^0 to L^0 .
- (c) What is the characteristic polynomial of σ acting on L^0 ?

Solution 8.

- (a) Since $|\text{Gal}(L/K)| = [L : K] = p$, $\text{Gal}(L/K)$ must be the cyclic group of order p . So, in particular, σ satisfies $x^p - 1 = 0$. Furthermore, σ in fact generates $\text{Gal}(L/K)$, so $\text{id}, \sigma, \sigma^2, \dots, \sigma^{p-1}$ are all the automorphisms of L/K . By linear independence of characters, these automorphisms are all linearly independent over K , so σ does not satisfy any polynomial of degree $< p$ over K . Hence, $x^p - 1$ is both the minimal and characteristic polynomial of σ .
- (b) Recall that the trace of an element is the sum of its Galois conjugates. So, if $\text{Tr}(\alpha) = 0$, then we have $\text{Tr}(\sigma(\alpha)) = \sum_{g \in \text{Gal}} g(\sigma(\alpha)) = \sum_{g' \in \text{Gal}} g'(\alpha) = 0$. Here we have used that the action of a group on itself by right multiplication is free.
- (c) Since $\text{Tr}(\alpha) = 0$ is a single K -linear condition, we know that L^0 is a dimension $p - 1$ vector space over K . So, we're looking for a degree $p - 1$ polynomial, and moreover we know that polynomial has to divide $x^p - 1$. Now, we have to split into cases based on the characteristic. If $\text{char}(K) = p$, then we know $x^p - 1 = (x - 1)^p$, and so if we're trying to find a degree $p - 1$ polynomial dividing it we have only one choice: $(x - 1)^{p-1}$.

If $\text{char}(K) \neq p$, we've gotta figure out which root of $x^p - 1$ is not an eigenvalue of σ acting on L^0 . And let's be honest, it'd be pretty weird if the answer wasn't the root 1, so let's prove that. If $\sigma(\alpha) = \alpha$, then $\text{Tr}(\alpha) = \sum_{i=0}^{p-1} \sigma^i(\alpha) = p\alpha = 0$. Since we are not in characteristic p , this implies $\alpha = 0$. Thus, σ has no eigenvectors of eigenvalue 1. Thus, the characteristic polynomial is $\frac{x^p - 1}{x - 1}$.